

Real-Time Time-Dependent Density Functional Theory for Periodic and Spin-Polarized Systems with the Real-Space Multigrids (RMG)[†]

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Abstract

[Abstract: to be written last, after all sections are complete. Approximately 200 words. Should mirror paper 1 abstract structure: (1) what we present, (2) what problem it solves, (3) key methodological contribution, (4) validation/benchmarking, (5) scalability demonstration and outlook.]

1 Introduction

Real-time time-dependent density functional theory (RT-TDDFT) has established itself as one of the primary tools for simulating the electronic response of materials to external electromagnetic perturbations.^{1,2} Its applications span a broad range of phenomena, including optical spectroscopy,³⁻⁷ charge transfer and photovoltaics,^{2,8,9} nanoplasmonics,¹⁰⁻¹² and spin and magnetization dynamics.¹³⁻¹⁵ In an earlier publication, we presented a scalable RT-TDDFT implementation within the Real-space MultiGrid (RMG) code for finite molecular systems, demonstrating simulations of up to 24,000 electrons on massively parallel architectures.¹⁶ The present work extends that implementation to periodic systems and spin-polarized calculations, significantly broadening the range of physically relevant problems accessible to RMG.

The transition from finite to periodic systems in RT-TDDFT requires a corresponding transition in the treatment of the electromagnetic perturbation. In finite systems, the length

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gauge—in which the coupling to an electric field is expressed through the dipole operator $\hat{\mathbf{r}}$ —is natural and convenient. In periodic systems, however, the position operator is ill-defined because it is not translationally invariant, and the dipole moment of an extended system is intrinsically multivalued.^{??} The correct framework for periodic systems is the velocity gauge, in which the electromagnetic interaction is expressed through the vector potential $\mathbf{A}(t)$ and the current operator, which is inherently translationally invariant. The transition between the two gauges is effected by a gauge transformation of the time-dependent Kohn-Sham equations,^{???} and requires careful treatment of the nonlocal part of pseudopotentials.^{???} The primary observable in the velocity gauge is the macroscopic current density $\mathbf{J}(t)$, from which the optical conductivity $\sigma(\omega)$ and dielectric function $\epsilon(\omega)$ are extracted by Fourier analysis.

RT-TDDFT for periodic systems has been implemented in several codes. The **Octopus** code,¹⁷ based on real-space grids, provides a comprehensive periodic RT-TDDFT implementation including both length and velocity gauges and has served as the primary validation reference for the present work. Plane-wave codes including **Qbox/Qball**,^{18,19} **cp2k**,²⁰ and **Exciting**[?] offer periodic RT-TDDFT with natural handling of Born-von Kármán boundary conditions. Atom-centered basis set codes such as **SIESTA**²¹ and **DFTB+**²² provide efficient implementations for systems where localized orbitals are appropriate. The **DFT-FE** code²³ offers an alternative real-space approach based on finite elements. [\[Check: add any missing periodic RT-TDDFT codes before submission. In particular: GPAW, Abinit, Yambo.\]](#)

The treatment of spin-polarized systems within RT-TDDFT introduces an additional degree of freedom: the electronic density is resolved into spin-up and spin-down components, and the time-dependent exchange-correlation potential depends on both. In the collinear approximation, the two spin channels evolve independently subject to a common self-consistent spin-dependent potential, and the optical response carries spin-specific signatures. This capability is essential for magnetic insulators such as CrI_3 , where the spin-up and spin-down optical spectra differ markedly and encode information about the electronic and magnetic structure.^{13,15} The non-collinear spin extension, in which the electronic states are represented as two-component spinors, is beyond the scope of the present work and will be addressed in a future publication.

A distinctive feature of the **RMG** implementation is its scalability. The real-space multi-grid approach assigns different regions of space to different processors, enabling near-linear scaling to hundreds or thousands of GPU-accelerated nodes.^{16,24} This scalability is particularly significant for periodic calculations with spin, where both the k-point sampling and the spin-polarized density matrix increase the computational cost relative to finite systems. To demonstrate this capability, we apply the periodic spin-polarized RT-TDDFT implementation to nitrogen-vacancy (NV) centers in diamond—a prototypical point defect system of substantial interest for quantum sensing and quantum information applications.[?] By systematically increasing the supercell size from [\[X\]](#) to [\[Y\]](#) atoms, we show how the optical response of the defect converges as the interaction between periodic images becomes negligible, identifying the minimum supercell size required for physically meaningful results at realistic defect concentrations.

The remainder of this paper is organized as follows. Section ?? develops the theoretical framework, beginning with the interaction of matter with electromagnetic radiation in the long-wavelength approximation, proceeding through the gauge transformation from length to velocity gauge, the definition of the current operator including nonlocal pseudopoten-

tial contributions, the density matrix propagation scheme, the spin-collinear extension, and the postprocessing route from current to optical conductivity. Section 3 describes the algorithm as implemented in **RMG** and identifies the key approximations and their physical justification. Section 4 presents benchmarking calculations for one-dimensional (H_2 chain), two-dimensional (hexagonal BN), and three-dimensional (Si) periodic systems, followed by spin-polarized calculations for CrI_3 , with comparisons to **Octopus** in all cases. The section concludes with the NV-center in diamond scalability demonstration. The paper concludes with a Summary and Outlook.

2 Theory and Methodology

2.1 Periodic RT-TDDFT and the Velocity Gauge

The starting point for the present periodic RT-TDDFT formulation is the time-dependent Kohn–Sham equation in the presence of an external electromagnetic perturbation,

$$i\hbar \frac{\partial}{\partial t} \Psi_j(\mathbf{r}, t) = \hat{H}(t) \Psi_j(\mathbf{r}, t), \quad (1)$$

where

$$\hat{H}(t) = \frac{1}{2m_e} \left[\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(\mathbf{r}, t) \right]^2 + q_e \Phi(\mathbf{r}, t) + \hat{V}_{\text{KS}}[\rho](t). \quad (2)$$

Here $q_e = -e$ is the electron charge, with $e > 0$, m_e is the electron mass, c is the speed of light, and

$$\hat{\mathbf{p}} = -i\hbar \nabla \quad (3)$$

is the canonical momentum operator. Thus, for an electron, the vector potential enters the kinetic momentum as $\hat{\mathbf{p}} + e\mathbf{A}/c$. In the norm-conserving pseudopotential formulation used here, the Kohn–Sham potential is decomposed as

$$\hat{V}_{\text{KS}}[\rho](t) = \hat{V}_{\text{ion}}^{\text{loc}} + \hat{V}_{\text{NL}} + \hat{V}_{\text{H}}[\rho](t) + \hat{V}_{\text{xc}}[\rho](t), \quad (4)$$

where $\hat{V}_{\text{ion}}^{\text{loc}}$ is the local ionic pseudopotential, $\hat{V}_{\text{NL}}$ is the nonlocal pseudopotential operator, $\hat{V}_{\text{H}}$ is the Hartree potential, and $\hat{V}_{\text{xc}}$ is the exchange-correlation potential.

The physical electric and magnetic fields are obtained from the electromagnetic potentials according to

$$\mathbf{E}(\mathbf{r}, t) = -\nabla \Phi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t}, \quad \mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (5)$$

Different choices of Φ and $\mathbf{A}$ related by a gauge transformation describe the same electromagnetic fields. This gauge freedom should be distinguished from the long-wavelength approximation used for optical response. In the present work we consider the homogeneous-field limit, in which the spatial variation of the external field over the simulation cell is neglected. This approximation corresponds to the optical, $\mathbf{q} \rightarrow 0$, limit and leads to vertical transitions at fixed crystal momentum. The $\mathbf{q} \rightarrow 0$ restriction therefore arises from the homogeneous-field approximation, not from the choice of velocity gauge.

For finite systems, a spatially uniform electric field is commonly represented in the length gauge by setting

$$\mathbf{A}_L(\mathbf{r}, t) = 0, \quad \Phi_L(\mathbf{r}, t) = -\mathbf{r} \cdot \mathbf{E}(t). \quad (6)$$

The corresponding scalar-potential energy for an electron is $q_e \Phi_L(\mathbf{r}, t) = e \mathbf{r} \cdot \mathbf{E}(t)$. This representation is well suited to finite molecular systems, where the time-dependent dipole moment is a well-defined observable. Under periodic boundary conditions, however, the position operator is not compatible with Born-von Karman periodicity: $\mathbf{r}$ and $\mathbf{r} + \mathbf{L}$ represent equivalent points of the crystal but give different values of the scalar potential in Eq. (6). The length-gauge scalar potential is therefore not a periodic operator.

The same homogeneous electric field can be represented in the velocity gauge by a gauge transformation. For a scalar gauge function $\chi(\mathbf{r}, t)$,

$$\mathbf{A}'(\mathbf{r}, t) = \mathbf{A}(\mathbf{r}, t) + \nabla \chi(\mathbf{r}, t), \quad \Phi'(\mathbf{r}, t) = \Phi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \chi(\mathbf{r}, t)}{\partial t}, \quad (7)$$

and the electronic wavefunction transforms as

$$\Psi'(\mathbf{r}, t) = \exp \left[\frac{ie}{\hbar c} \chi(\mathbf{r}, t) \right] \Psi(\mathbf{r}, t). \quad (8)$$

Starting from the length-gauge potentials in Eq. (6), choose

$$\chi(\mathbf{r}, t) = \mathbf{r} \cdot \mathbf{A}(t), \quad \frac{d\mathbf{A}(t)}{dt} = -c\mathbf{E}(t). \quad (9)$$

This gives

$$\Phi_V(\mathbf{r}, t) = 0, \quad \mathbf{A}_V(\mathbf{r}, t) = \mathbf{A}(t), \quad \mathbf{E}(t) = -\frac{1}{c} \frac{d\mathbf{A}(t)}{dt}. \quad (10)$$

For an electron, the corresponding unitary transformation is

$$\hat{U}(t) = \exp \left[-\frac{ie}{\hbar c} \mathbf{r} \cdot \mathbf{A}(t) \right], \quad \Psi_V(\mathbf{r}, t) = \hat{U}(t) \Psi_L(\mathbf{r}, t). \quad (11)$$

Because $\mathbf{A}(t)$ is spatially uniform, $\nabla \times \mathbf{A}(t) = 0$; the homogeneous velocity-gauge field is therefore the gauge-equivalent representation of the electric-dipole field, not a full finite- $\mathbf{q}$ transverse electromagnetic wave. Finite- $\mathbf{q}$ response functions lie outside the homogeneous-field formulation considered here.

This relation also fixes the formal connection between a conventional broadband electric-field kick and the corresponding velocity-gauge vector potential. For an impulsive perturbation,

$$\mathbf{E}(t) = \mathbf{E}_0 \delta(t), \quad (12)$$

integration of Eq. (10) gives, for $\mathbf{A}(t < 0) = 0$ under the sign conventions established above,

$$\mathbf{A}(t) = -c\mathbf{E}_0 \Theta(t). \quad (13)$$

The particular impulsive realization used in RMG is described separately in Sec. 3.

In the formal velocity-gauge representation, the periodic TDKS Hamiltonian for an elec-

tron is

$$\hat{H}_V(t) = \frac{1}{2m_e} \left[\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(t) \right]^2 + \hat{V}_{\text{loc}}[\rho](t) + \hat{V}_{\text{NL}}^V(t). \quad (14)$$

Here

$$\hat{V}_{\text{loc}}[\rho](t) = \hat{V}_{\text{ion}}^{\text{loc}} + \hat{V}_{\text{H}}[\rho](t) + \hat{V}_{\text{xc}}[\rho](t) \quad (15)$$

contains all local terms, while the nonlocal pseudopotential is written separately. Since $q_e = -e$, the kinetic momentum in Eq. (14) is $\hat{\mathbf{p}} + e\mathbf{A}(t)/c$ for an electron. If a nonlocal pseudopotential is present, it must be transformed consistently with the same unitary gauge transformation,

$$\hat{V}_{\text{NL}}^V(t) = \hat{U}(t) \hat{V}_{\text{NL}} \hat{U}^\dagger(t), \quad \hat{U}(t) = \exp \left[\frac{iq_e}{\hbar c} \mathbf{r} \cdot \mathbf{A}(t) \right]. \quad (16)$$

The detailed consequences of Eq. (16) for the current operator are discussed in Sec. 2.2.

For a periodic system, the time-dependent Kohn–Sham orbitals are written in Bloch form,

$$\Psi_{n\mathbf{k}}(\mathbf{r}, t) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}, t), \quad (17)$$

where $u_{n\mathbf{k}}(\mathbf{r}, t)$ has the periodicity of the unit cell. Acting on a Bloch orbital, the canonical momentum becomes

$$\hat{\mathbf{p}} \Psi_{n\mathbf{k}} = e^{i\mathbf{k} \cdot \mathbf{r}} (-i\hbar \nabla + \hbar \mathbf{k}) u_{n\mathbf{k}}. \quad (18)$$

The corresponding cell-periodic equation is

$$i\hbar \frac{\partial}{\partial t} u_{n\mathbf{k}}(\mathbf{r}, t) = \hat{H}_{\mathbf{k}}^V(t) u_{n\mathbf{k}}(\mathbf{r}, t), \quad (19)$$

with

$$\hat{H}_{\mathbf{k}}^V(t) = \frac{1}{2m_e} \left[-i\hbar \nabla + \hbar \mathbf{k} - \frac{q_e}{c} \mathbf{A}(t) \right]^2 + \hat{V}_{\text{loc}}[\rho](t) + \hat{V}_{\text{NL},\mathbf{k}}^V(t). \quad (20)$$

Equivalently, using $q_e = -e$, the kinetic operator contains $-i\hbar \nabla + \hbar \mathbf{k} + e\mathbf{A}(t)/c$. Equation (20) is the periodic counterpart of the finite-system length-gauge RT-TDDFT formulation used in our previous work,¹⁶ but with the position-dependent dipole perturbation replaced by a periodic velocity-gauge coupling.

Because the dipole moment is not a well-defined bulk observable under periodic boundary conditions, the natural time-dependent observable for periodic optical response is the macroscopic current rather than the molecular dipole moment. The construction of the current operator, including the nonlocal pseudopotential contribution, is given in Sec. 2.2; the extraction of optical response functions from the real-time current is described in Sec. ??.

2.2 Current Operator and Nonlocal Pseudopotentials

For a periodic system in the homogeneous-field limit, the optical response is naturally formulated in terms of the macroscopic charge-current density. We define

$$J_\alpha(t) = \frac{1}{V_{\text{cell}}} \sum_{\mathbf{k}, \sigma} w_{\mathbf{k}} \text{Tr} \left[\mathbf{P}_{\mathbf{k}}^\sigma(t) \hat{J}_{\mathbf{k}, \alpha}^\sigma(t) \right], \quad (21)$$

where V_{cell} is the volume of the periodic simulation cell, which may be a primitive cell or a supercell, $w_{\mathbf{k}}$ are the normalized Brillouin-zone weights satisfying $\sum_{\mathbf{k}} w_{\mathbf{k}} = 1$, $\mathbf{P}_{\mathbf{k}}^{\sigma}$ is the single-particle density matrix for spin channel σ , and

$$\hat{I}_{\alpha}(t) = q_e \hat{v}_{\alpha}(t) \quad (22)$$

is the charge-current operator. The trace in Eq. (21) is taken over the propagated single-particle subspace at fixed $\mathbf{k}$ and spin.

The velocity operator is most cleanly defined before introducing the cell-periodic Bloch representation. For the velocity-gauge Hamiltonian of Eq. (14),

$$\hat{v}_{\alpha}(t) = \frac{i}{\hbar} \left[\hat{H}_V(t), \hat{r}_{\alpha} \right]. \quad (23)$$

Since the local potential commutes with the position operator, the Hamiltonian containing a local potential and a nonlocal pseudopotential gives

$$\hat{v}_{\alpha}(t) = \frac{1}{m_e} \left[\hat{p}_{\alpha} - \frac{q_e}{c} A_{\alpha}(t) \right] + \frac{i}{\hbar} \left[\hat{V}_{\text{NL}}^V(t), \hat{r}_{\alpha} \right]. \quad (24)$$

Equivalently,

$$\frac{i}{\hbar} \left[\hat{V}_{\text{NL}}^V(t), \hat{r}_{\alpha} \right] = \frac{1}{i\hbar} \left[\hat{r}_{\alpha}, \hat{V}_{\text{NL}}^V(t) \right], \quad (25)$$

which fixes the sign convention for the nonlocal contribution. Thus the formal charge-current operator is

$$\hat{I}_{\alpha}(t) = q_e \left\{ \frac{1}{m_e} \left[\hat{p}_{\alpha} - \frac{q_e}{c} A_{\alpha}(t) \right] + \frac{i}{\hbar} \left[\hat{V}_{\text{NL}}^V(t), \hat{r}_{\alpha} \right] \right\}. \quad (26)$$

The explicit $A_{\alpha}(t)$ dependence of the kinetic term gives the usual diamagnetic contribution,

$$\hat{I}_{\alpha}^{\text{dia}}(t) = -\frac{q_e^2}{m_e c} A_{\alpha}(t) \hat{1}, \quad (27)$$

while the nonlocal term is also A dependent through the gauge-transformed nonlocal pseudopotential.

The same nonlocal commutator appears when the velocity-gauge Hamiltonian is expanded to first order in the homogeneous vector potential. Using Eq. (16),

$$\hat{V}_{\text{NL}}^V(t) = \hat{U}(t) \hat{V}_{\text{NL}} \hat{U}^{\dagger}(t), \quad \hat{U}(t) = \exp \left[\frac{iq_e}{\hbar c} \mathbf{r} \cdot \mathbf{A}(t) \right], \quad (28)$$

one obtains

$$\hat{V}_{\text{NL}}^V(t) = \hat{V}_{\text{NL}} + \frac{iq_e}{\hbar c} \sum_{\alpha} A_{\alpha}(t) \left[\hat{r}_{\alpha}, \hat{V}_{\text{NL}} \right] + \mathcal{O}(A^2). \quad (29)$$

The first-order nonlocal perturbation is therefore

$$\delta \hat{V}_{\text{NL}}^{(1)}(t) = -\frac{1}{c} \sum_{\alpha} A_{\alpha}(t) \hat{I}_{\alpha, \text{NL}}^{(0)}, \quad \hat{I}_{\alpha, \text{NL}}^{(0)} = \frac{q_e}{i\hbar} \left[\hat{r}_{\alpha}, \hat{V}_{\text{NL}} \right]. \quad (30)$$

Combining this term with the kinetic part of the minimal-coupling Hamiltonian gives the linear velocity-gauge perturbation

$$\delta\hat{H}^{(1)}(t) = -\frac{1}{c} \sum_{\alpha} A_{\alpha}(t) \hat{I}_{\alpha}^{(0)}, \quad (31)$$

where

$$\hat{I}_{\alpha}^{(0)} = q_e \left[\frac{\hat{p}_{\alpha}}{m_e} + \frac{i}{\hbar} [\hat{V}_{\text{NL}}, \hat{r}_{\alpha}] \right] = \frac{q_e}{m_e} \hat{p}_{\alpha} + \frac{q_e}{i\hbar} [\hat{r}_{\alpha}, \hat{V}_{\text{NL}}]. \quad (32)$$

Equations (26)–(32) show that the nonlocal-pseudopotential commutator is not only a correction to the measured current; it is also the nonlocal part of the linear velocity-gauge perturbation required for gauge consistency when nonlocal pseudopotentials are present.²⁵ More generally, a nonlocal one-particle operator present in the Hamiltonian contributes to the velocity through the corresponding commutator with the position operator.

For a separable norm-conserving pseudopotential,²⁶

$$\hat{V}_{\text{NL}} = \sum_{I,lm} |\beta_{I,lm}\rangle D_{I,lm} \langle\beta_{I,lm}|, \quad (33)$$

the coordinate-space matrix element of the commutator can be written as

$$\langle \mathbf{r} | [\hat{r}_{\alpha}, \hat{V}_{\text{NL}}] | \mathbf{r}' \rangle = (r_{\alpha} - r'_{\alpha}) V_{\text{NL}}(\mathbf{r}, \mathbf{r}'). \quad (34)$$

Equation (34) makes explicit that this term vanishes for a strictly local potential but is finite for a nonlocal pseudopotential.

The corresponding cell-periodic Bloch representation is obtained only after the full-space current operator has been defined. For $\Psi_{n\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}$, the canonical momentum becomes $-i\hbar\nabla + \hbar\mathbf{k}$ as in Eq. (18). The nonlocal operator at fixed $\mathbf{k}$ is

$$\hat{V}_{\text{NL},\mathbf{k}} = e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{V}_{\text{NL}} e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (35)$$

At zero external vector potential, the charge-current matrix in a cell-periodic basis is therefore

$$\begin{aligned} I_{ab,\mathbf{k},\alpha}^{(0),\sigma} &= \frac{q_e}{m_e} \langle u_{a\mathbf{k}}^{\sigma} | -i\hbar\nabla_{\alpha} + \hbar k_{\alpha} | u_{b\mathbf{k}}^{\sigma} \rangle_{\text{cell}} \\ &+ \frac{q_e}{i\hbar} \left\langle u_{a\mathbf{k}}^{\sigma} \left| [\hat{r}_{\alpha}, \hat{V}_{\text{NL},\mathbf{k}}^{\sigma}] \right| u_{b\mathbf{k}}^{\sigma} \right\rangle_{\text{cell}}. \end{aligned} \quad (36)$$

The first term contains both the gradient and Bloch $\hbar k_{\alpha}$ contributions, while the second term is the nonlocal-pseudopotential contribution. In the presence of a finite homogeneous vector potential, Eq. (26) gives the corresponding A -dependent current operator, including the diamagnetic kinetic term and the gauge-transformed nonlocal operator.

Equations (21) and (36) define the formal current observable used in the periodic RT-TDDFT theory. The implementation-specific construction of the current matrices, the particular impulsive realization of the velocity-gauge perturbation, and the conversion of raw current traces to optical response functions are described separately in Secs. 3 and ??.

3 Implementation

The implementation of the periodic and spin-polarized RT-TDDFT presented in this work is built on the existing finite-system RT-TDDFT module of **RMG**,¹⁶ which is described in detail in the accompanying paper. Here we focus on the extensions required for periodic boundary conditions and spin-polarized calculations, the key approximations and their physical justification, and the computational organization of the velocity gauge propagation.

3.1 Algorithm

The computational scheme for periodic RT-TDDFT in **RMG** is summarized in Algorithm 1. It follows the same density matrix propagation framework as the finite-system implementation¹⁶ but with three key modifications: (1) the perturbation is applied as a momentum kick via the vector potential $\mathbf{A}$ rather than an electric field dipole kick; (2) the current operator matrix replaces the dipole matrix as the primary observable; and (3) the Hamiltonian matrix is evaluated at each k-point over the Brillouin zone.

Algorithm 1 Periodic RT-TDDFT Algorithm (VECTOR_POT mode)

```

/* Ground state DFT calculations: */
1: Generate Bloch orbitals  $|\psi_{n,\mathbf{k}}\rangle$ , Kohn-Sham Hamiltonian  $\mathbf{H}_{0,\mathbf{k}}^\sigma$  for each k-point  $\mathbf{k}$  and spin  $\sigma$ 
2: Select active space:  $N_{\text{occ}}$  occupied +  $N_{\text{virt}}$  virtual orbitals

/* Build current operator matrices (once, before time loop): */
3: For each k-point  $\mathbf{k}$  and spin  $\sigma$ :
4:   Build paramagnetic current matrix:
       $J_{ab}^\alpha \leftarrow \langle \psi_a | (-i\nabla_\alpha + k_\alpha) | \psi_b \rangle$  [VecPHmatrix()]
5:   Add nonlocal PP correction:
       $J_{ab}^\alpha \leftarrow J_{ab}^\alpha + \langle \psi_a | [\hat{r}_\alpha, V^{\text{nl}}] | \psi_b \rangle$  [CurrentNlpp()]
6:   Add  $\mathbf{A}(t_0) \cdot \mathbf{J}$  to  $\mathbf{H}_{-1,\mathbf{k}}$  to encode the perturbation kick at  $t = 0$ 

/* Initialize density matrix: */
7:  $\mathbf{P}_\mathbf{k}^0 \leftarrow \text{diag}(f_{n,\mathbf{k}})$  (occupation numbers on diagonal)
8:  $\mathbf{H}_{-1,\mathbf{k}} \leftarrow \mathbf{H}_{0,\mathbf{k}}$  (extrapolation seed)

/* Time loop: */
9: for  $j = 1$  to  $N_{\text{steps}}$  do
10:    $t \leftarrow t_0 + j \cdot \Delta t$ 
11:   Extrapolate:  $\mathbf{H}_{1,\mathbf{k}} \leftarrow 2\mathbf{H}_{0,\mathbf{k}} - \mathbf{H}_{-1,\mathbf{k}}$  [extrapolate_Hmatrix()]

/* Self-consistent Hamiltonian loop: */
12:   while  $\|\mathbf{H}_1^{\text{new}} - \mathbf{H}_1^{\text{old}}\|_\infty > H_{\text{thrs}}$  do
13:     Magnus operator:  $\Omega_\mathbf{k} \leftarrow -i \cdot \frac{1}{2} (\mathbf{H}_{0,\mathbf{k}} + \mathbf{H}_{1,\mathbf{k}}) \Delta t$  [magnus()]
14:     Propagate density matrix:  $\mathbf{P}_\mathbf{k}^1 \leftarrow \mathbf{P}_\mathbf{k}^0 + [\Omega_\mathbf{k}, \mathbf{P}_\mathbf{k}^0] + \dots$  [eldyn_ort() / commutp()]
15:     Update density:  $\rho^\sigma(\mathbf{r}) \leftarrow \rho_{\text{gs}}^\sigma(\mathbf{r}) + \sum_\mathbf{k} w_\mathbf{k} \sum_n P_{nn,\mathbf{k}}^1 |\psi_{n,\mathbf{k}}(\mathbf{r})|^2$  [GetNewRho_rmgtddft()]
16:     Update potentials:  $V_{\text{H}}[\rho], V_{\text{xc}}^\sigma[\rho^\uparrow, \rho^\downarrow]$ 
17:     Update Hamiltonian:  $\mathbf{H}_{1,\mathbf{k}}^{\text{new}} \leftarrow \langle \psi_a | \hat{H}[\rho] | \psi_b \rangle$  [HmatrixUpdate()]
18:   end while

19:   Extract current:  $J_\alpha(t) \leftarrow \text{Re} \sum_\mathbf{k} w_\mathbf{k} \text{Tr}[\mathbf{P}_\mathbf{k}^1 \cdot \mathbf{J}_\mathbf{k}^\alpha]$ 
20:   (Optional) Berry phase polarization:  $P_{\text{BP},\alpha}(t) \leftarrow \text{Tr}[\mathbf{P}_\mathbf{k}^1 \cdot \mathbf{X}_\mathbf{k}^\alpha]$ 
21:   Advance:  $\mathbf{P}^0 \leftarrow \mathbf{P}^1$ ;  $\mathbf{H}_{-1} \leftarrow \mathbf{H}_0$ ;  $\mathbf{H}_0 \leftarrow \mathbf{H}_1$ 
22: end for

/* Postprocessing: */
23: Fourier transform  $J_\alpha(t)$  for each Cartesian direction  $\alpha$ 
24: Compute  $\sigma_{\alpha\beta}(\omega), \varepsilon_{\alpha\beta}(\omega)$ 

```

3.2 Key Approximations and Their Justification

The implementation described in Algorithm 1 embodies several approximations with respect to the full theory presented in Section ???. We discuss each in turn.

Perturbative momentum kick. The full velocity gauge Hamiltonian (eq. ??) contains the vector potential $\mathbf{A}(t)$ at every time step, requiring the $\mathbf{A}(t) \cdot \hat{\mathbf{p}}$ term to be re-evaluated throughout the propagation. In the present implementation, we instead apply $\mathbf{A}(t)$ as a delta-function kick at $t = 0$ only: the momentum kick is encoded in the initial Hamiltonian $\mathbf{H}_{-1}$ by adding the $\mathbf{A}(t_0) \cdot \mathbf{J}^\alpha$ matrix (step 6 in Algorithm 1), after which $\mathbf{A} = 0$ for all $t > 0$. This approach is equivalent to first-order perturbation theory in $|\mathbf{A}_0|$, and it provides the full linear optical response of the system through the free oscillations of the current density $\mathbf{J}(t)$ following the kick. The linear response regime is valid as long as the kick amplitude is small enough that the system does not leave the linear regime, which is straightforwardly verified by checking that the response is proportional to $|\mathbf{A}_0|$.

Diamagnetic current term. The full current density contains both paramagnetic and diamagnetic contributions (Section ??). In the present implementation, the diamagnetic term $\mathbf{J}_{\text{dia}} \propto \mathbf{A}(t)|\psi|^2$ is identically zero for all $t > 0$ because $\mathbf{A}(t > 0) = 0$ in the perturbative kick scheme. This is consistent with the linear response regime: the diamagnetic contribution to the linear response function is implicitly accounted for by the sum rule, and the paramagnetic current alone gives the correct optical conductivity.^{??} Simulations with a continuously applied driving field (e.g., a CW laser or a Gaussian pulse) would require retaining $\mathbf{A}(t)$ throughout the propagation and are beyond the scope of the present work.

First-order Magnus expansion. As in the finite-system implementation,¹⁶ the Magnus operator is truncated to first order:

$$\Omega \approx \Omega_1 = -i \cdot \frac{1}{2}(\mathbf{H}_0 + \mathbf{H}_1)\Delta t. \quad (37)$$

The second-order correction $\Omega_2 = \frac{1}{12}\Delta t^2[\mathbf{H}_0, \mathbf{H}_1]$ is not included. The accuracy and stability of this scheme for the time steps used here was validated in paper 1 through energy conservation tests, and we demonstrate analogous stability for the periodic case in Section 4.

Norm-conserving pseudopotentials only. The current implementation supports only norm-conserving pseudopotentials (NCPP).^{??} Ultrasoft pseudopotentials (USPP)²⁷ introduce a non-trivial overlap matrix $\mathbf{S} \neq \mathbf{I}$ into the density matrix propagation (the nonorthogonal basis path), which requires the generalized propagation scheme described in Appendix A of paper 1. This extension is planned for future work.

BCH propagator for complex Hamiltonians. For k-points away from the Γ point, the Hamiltonian matrix $\mathbf{H}_{\mathbf{k}}$ is complex Hermitian. In this case, the BCH commutator expansion propagator (Ieldyn=1) is used throughout; the diagonalization-based propagator (Ieldyn=2) is not currently supported for complex matrices. The BCH expansion is unconditionally

stable and has been shown to give results equivalent to exact diagonalization within the convergence threshold,¹⁶ and we find no practical limitation from this choice.

3.3 Software Optimization and Parallelization

[Content outline – adapt from paper 1 Section 3.2 with] additions for the periodic case:
 - Same MPI + OpenMP + GPU parallelization as paper 1 - k-point parallelization: each MPI group handles a subset of k-points (pct.kpsub_comm); current is summed over k-points with weights (MPI_Allreduce over kpsub_comm, lines 828-829 in RmgTddft.cpp) - Spin parallelization: separate MPI communicator groups per spin channel (pct.spin_comm); spin channels communicate only for XC update (MPI_Allreduce over spin_comm, line 767) - The density matrix P and Hamiltonian H are distributed using ScaLAPACK layout (Sp object); size is numst x numst where numst = N_occ + N_virt - For VECTOR_POT mode: momentum matrix J_alpha is complex even at Gamma point (stored as complex<double> in Pxmatrix_cpu etc.) - The BCH commutator $[\Omega, P]$ dominates $O(N^3)$; same optimization as paper 1 (real/imaginary decomposition, BLAS ZGEMM/DGEMM) - Timing data for benchmark systems: [add when available]

[Note: include a brief timing comparison table analogous to Table 1 in paper 1, but for one or more periodic benchmark systems (e.g., Si supercells of increasing size) to demonstrate scaling.]

4 Numerical Results

We benchmark the periodic and spin-polarized RT-TDDFT implementation in **RMG** against the **Octopus** code¹⁷ for four representative systems of increasing dimensionality and physical complexity: a one-dimensional hydrogen chain (H_2), a two-dimensional hexagonal boron nitride (hBN) monolayer, a three-dimensional silicon crystal, and the spin-polarized magnetic insulator CrI_3 . These systems are chosen to isolate and validate specific aspects of the implementation: k-point sampling, anisotropic optical response, three-dimensional periodic boundaries, and spin-collinear propagation, respectively. The section concludes with a demonstration of the unique large-scale capability of **RMG** through calculations on nitrogen-vacancy (NV) centers in diamond at varying supercell sizes.

In all calculations, a momentum kick of amplitude $E_0 = [\text{value}]$ a.u. was applied at $t = 0$ along each Cartesian direction independently, and the optical conductivity was obtained from the Fourier transform of the resulting current density. [Add: common computational parameters – time step, total propagation time, broadening gamma, XC functional, pseudopotential library used for all systems.]

4.1 One-Dimensional Periodic System: Hydrogen Chain

As the simplest possible periodic test system, we consider an infinite chain of hydrogen molecules aligned along the z axis with alternating bond lengths, modeled by a one-dimensional supercell with periodic boundary conditions. This system provides a stringent test of k-point

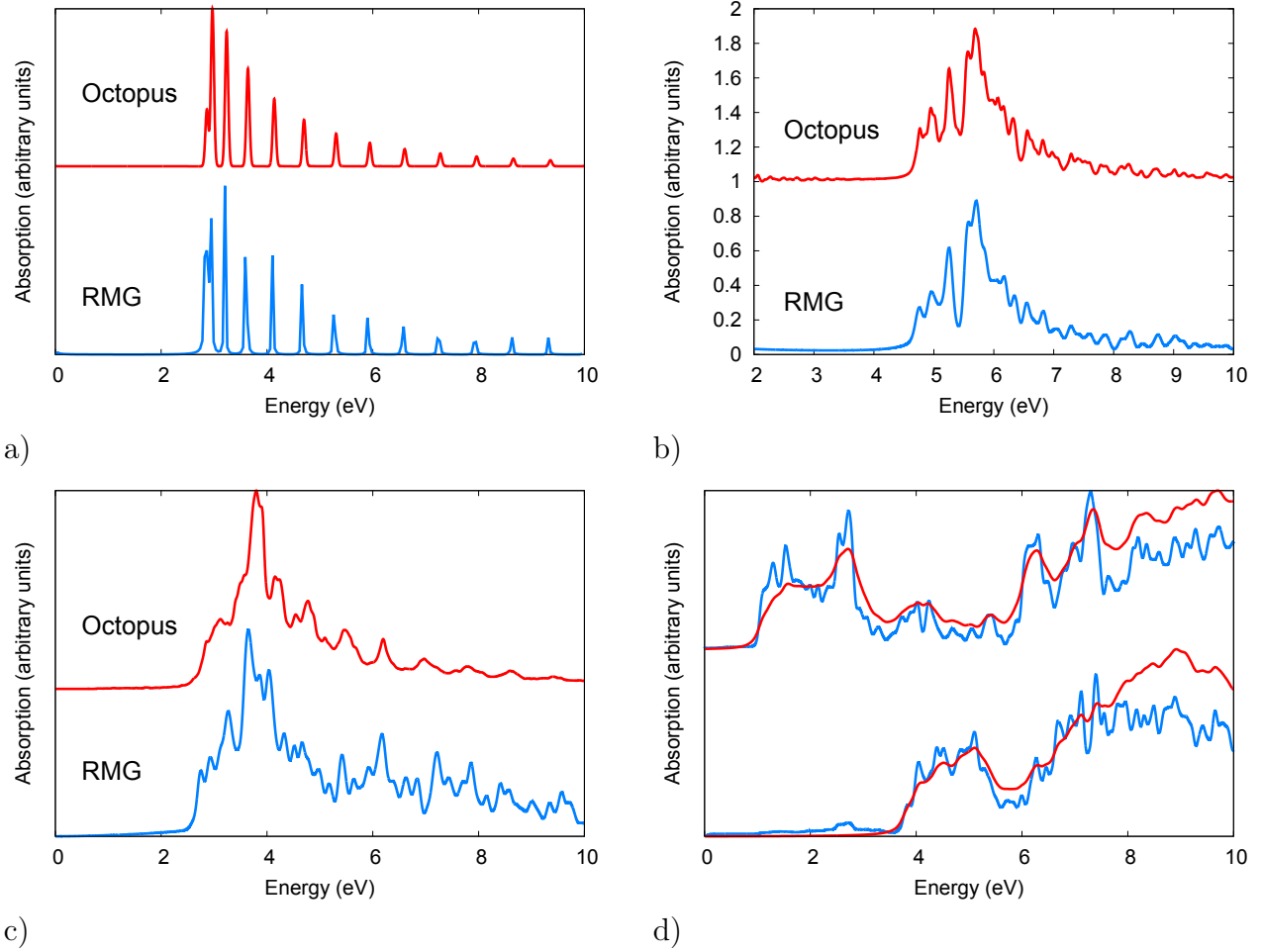


Figure 1: Optical conductivity of (a) 1-D Hydrogen, (b) 2-D hexagonal Boron Nitride (c) 3-D Silicon, (d) spin polarized CrI3 from RMG compared with Octopus

sampling and the velocity gauge implementation in the limit of a quasi-one-dimensional geometry, where the longitudinal (z -direction) and transverse (x, y -direction) optical responses differ substantially.

[Computational details: - Supercell geometry: H-H bond length, H₂-H₂ separation - K-point sampling: [N_k points along z , Gamma in x and y] - Grid spacing: [value] a_0 - XC functional: [PBE or LDA] - Pseudopotential: [type] - Time step Δt : [value] a.u. - Total simulation time: [value] a.u. - Broadening γ : [value] eV]

[Discussion points: - Agreement between RMG and Octopus for both longitudinal and transverse components - K-point convergence: how many k-points needed for converged spectrum? - Physical interpretation of the spectral features - If Berry phase comparison is available: show agreement between current-integration and BP polarization as internal consistency check (Fruit 1 from CONTEXT.md) - If ground-state current convergence is shown: include as panel or note in text (Fruit 2 from CONTEXT.md)]

4.2 Two-Dimensional Periodic System: Hexagonal Boron Nitride

Hexagonal boron nitride (hBN) is a wide-gap insulating two-dimensional material with D_{6h} symmetry. Its optical response is highly anisotropic: the in-plane conductivity $\sigma_{xx}(\omega) = \sigma_{yy}(\omega)$ differs markedly from the out-of-plane component $\sigma_{zz}(\omega)$, the latter being negligible for photon energies below the interlayer excitations. This system tests the implementation of two-dimensional periodic boundary conditions and the symmetry enforcement of the conductivity tensor in RMG.

[Computational details: - Monolayer geometry: lattice constant [a] Å, vacuum layer [d] Å to prevent inter-layer interactions - K-point sampling: [N_k × N_k × 1] Monkhorst-Pack - Grid spacing: [value] a₀ - XC functional: [PBE] - Pseudopotential: [type] - Time step and total simulation time: [values]]

[Discussion points: - Agreement between RMG and Octopus for in-plane conductivity - Symmetry: show that $\sigma_{xx} = \sigma_{yy}$ (Fruit 3 from CONTEXT.md) - Physical features: onset at band gap, first excitonic peak if visible - Comparison with available experimental or LR-TDDFT data if relevant]

4.3 Three-Dimensional Periodic System: Silicon

Silicon in the diamond cubic structure ($Fd\bar{3}m$, space group 227) is the canonical benchmark for periodic electronic structure calculations. Its indirect band gap and well-characterized optical spectrum provide a rigorous test of the full three-dimensional periodic RT-TDDFT implementation. The cubic symmetry requires $\sigma_{xx}(\omega) = \sigma_{yy}(\omega) = \sigma_{zz}(\omega)$, providing an immediate symmetry validation.

[Computational details: - Unit cell: conventional 8-atom cell, lattice constant [a] Å - K-point sampling: [N_k × N_k × N_k] Monkhorst-Pack - Grid spacing: [value] a₀ - XC functional: [PBE] - Pseudopotential: [type, NCPP] - Number of occupied + virtual states: [N_occ + N_virt] - Time step Δt : [value] a.u. - Total simulation time: [value] a.u. ([value] fs) - Broadening γ : [value] eV]

[Discussion points: - Agreement between RMG and Octopus: peak positions, lineshapes - Symmetry: $\sigma_{xx} = \sigma_{yy} = \sigma_{zz}$ (Fruit 3) - Energy conservation test for periodic case (Fruit 5 from CONTEXT.md): run longer simulation and show no drift, analogous to paper 1 Fig. 3c - Berry phase comparison if included (Fruit 1) - Physical features: onset near direct gap, main optical peak structure]

4.4 Spin-Polarized System: CrI₃

Chromium triiodide (CrI₃) is a layered van der Waals magnetic insulator with a honeycomb lattice of Cr³⁺ ions, each carrying a local moment of approximately $3\mu_B$. In its ferromagnetic ground state, the spin-up and spin-down electronic structures differ, leading to distinct optical spectra for the two spin channels. This system provides a direct test of the spin-collinear RT-TDDFT extension: the two spin channels should propagate independently but with a common Hartree potential, yielding spin-resolved optical conductivities that can be compared with Octopus and with available experimental data.

[Computational details: - Unit cell: [describe geometry and magnetic ordering] - K-point sampling: $[N_k \times N_k \times N_k]$ - XC functional: [PBE or PBE+U; specify U if used] - Pseudopotential: [type, NCPP] - Initial spin configuration: ferromagnetic, moments on Cr sites - Time step and total simulation time: [values] - Broadening: [value] eV]

[Discussion points: - Agreement between RMG and Octopus for both spin channels - Physical interpretation: which spectral features are spin-up vs spin-down? Charge-transfer excitations, Cr d-d transitions - Spin current $\mathbf{J}_{\text{spin}} = \mathbf{J}^{\uparrow} - \mathbf{J}^{\downarrow}$ if computed (Fruit 4 from CONTEXT.md) - Comparison with available experimental optical data if relevant - Note: ECD/MCD spectra not computed in this work (magnetic perturbation formalism reserved for future publication)]

4.5 JACEK: Large-Scale Demonstration: NV Center in Diamond

As a final demonstration, we apply the periodic spin-polarized RT-TDDFT implementation to the negatively charged nitrogen-vacancy (NV^-) center in diamond, one of the most studied point defects in solid-state physics.^{28–30} The ground state of NV^- center is an optically addressable spin-triplet located within the diamond band gap^{29–31} Because of its optical addressability and long coherence times at room temperature the NV^- center is an attractive system for several emerging quantum technologies, including single-photon emitter, nanoscale magnetometry, quantum sensing, quantum information processing.^{32–37}

This system tests several capabilities simultaneously: spin-polarized real-time propagation, optical response of a localized defect in a three-dimensional periodic host, convergence toward the dilute-defect limit, as well as computational scaling to thousands of atoms. The purpose of this section is not to provide a complete quantitative description of NV^- photophysics. Instead, the present calculations examine the electronic optical response of defect in the energy region of the experimental zero-phonon line (ZPL) and use the NV^- center to demonstrate how polarization selection rules, periodic defect-image interactions, and active-space truncation affect large-scale periodic RT-TDDFT calculations.

4.5.1 Electronic structure and simulation setup for NV^-

The NV^- center consists of a substitutional nitrogen atom adjacent to a carbon vacancy, with one additional electron trapped at the defect (Figure 2a). The defect axis is oriented along the crystallographic $[111]$ direction, which defines the C_3 symmetry axis of the approximately C_{3v} local defect environment. The defect states are located within the diamond band gap. The lower a_1 -like defect orbital has substantial nitrogen character together with contributions from the three carbon atoms adjacent to the vacancy, whereas the nearly degenerate e manifold is primarily localized on these three carbon atoms, as shown in Figure 2c–e. Because pristine diamond has a wide band gap of about 5.5 eV and is transparent throughout the visible range, optical absorption and emission below this energy arise from defect states; for NV^- , the ${}^3A_2 \rightarrow {}^3E$ transition has an experimental zero-phonon line at 1.945 eV, corresponding to 637 nm red light.

All NV^- calculations were performed here with the RMG code using periodic boundary conditions, the PBE exchange-correlation functional, and the NC_accuracy norm-conserving pseudopotentials. The diamond lattice constant was first optimized for bulk diamond using

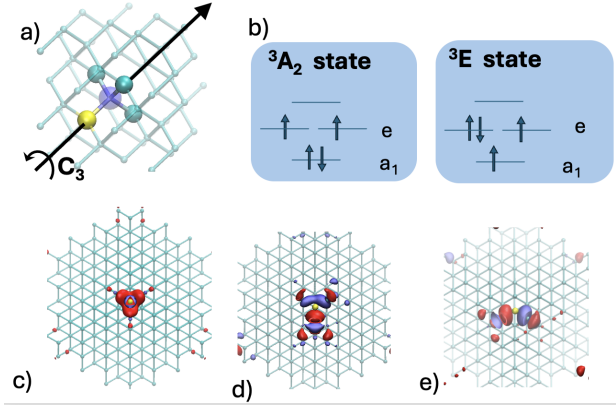


Figure 2: Electronic structure of the negatively charged nitrogen-vacancy (NV^-) center in diamond. (a) Local atomic structure of the defect, consisting of a substitutional nitrogen atom (green) adjacent to a carbon vacancy (blue transparent). The black arrow indicates the $[111]$ direction, which defines the C_3 symmetry axis of the approximately C_{3v} defect environment. (b) Schematic occupation of the defect levels in the triplet ground state, 3A_2 (configuration $a_1^2 e_1^1 e_2^1$), and in the optically excited triplet 3E state (configuration $a_1^1 e_1^2 e_2^1$). The defect-associated optical transition in the experimental zero-phonon-line energy region is dominated by promotion of a minority-spin electron from the occupied a_1 -like defect orbital into the e manifold. (c–e) Representative real-space isosurfaces of Kohn–Sham defect orbitals localized in the diamond band gap (view along the C_3 axis: (c) the occupied a_1 -like orbital and (d,e) the two components of the nearly degenerate e manifold. Red and blue isosurfaces denote opposite phases of the Kohn–Sham orbitals.

the same functional and pseudopotentials, giving $a = 3.5725125 \text{ \AA}$. This optimized lattice constant was then kept fixed for all NV^- defect calculations.

Three single-defect supercell sizes were considered, denoted NV- $x2$ (small), NV- $x4$ (medium), and NV- $x8$ (large). These systems were constructed from $2 \times 2 \times 2$, $4 \times 4 \times 4$, and $8 \times 8 \times 8$ repetitions of the conventional face-centered cubic (fcc) diamond cell, corresponding to 64, 512, and 4096 diamond lattice sites before formation of the defect. Previous supercell studies of NV^- in diamond have shown that relatively large cells, on the order of the 512-site NV- $x4$ model, are needed to reduce spurious interactions between periodic defect images.^{29–31} Here we directly test this finite-size behavior by extending the real-time calculations to the 4096-site NV- $x8$ cell. The same 4096-site host also enables explicit simulations of periodic arrays of NV^- centers, including collinear parallel and antiparallel defect-spin arrangements, which are used below as an additional demonstration of spin-resolved periodic RT-TDDFT. The actual defect supercells contained 63, 511, and 4095 atoms, respectively. The negatively charged supercells were treated using a uniform compensating background charge to maintain overall charge neutrality under periodic boundary conditions.

The real-space grid was scaled with the supercell size so that all three calculations used the same grid spacing. The NV- $x2$, NV- $x4$, and NV- $x8$ systems used 48^3 , 96^3 , and 192^3 grid points, respectively, corresponding to a grid spacing of $0.281 a_0$. This grid corresponds to equivalent wavefunction and charge-density cutoffs of 124.7 and 498.9 Ry, respectively; the charge-density grid is two times finer in each direction.

After construction of the charged spin-polarized defect cells, all atomic coordinates were relaxed at fixed lattice vectors. For each supercell size, the spin-polarized triplet solution was found to be lower in energy than the corresponding singlet solution, and the relaxed triplet

geometries were used as the starting point for the RT-TDDFT simulations. All real-time calculations used spin-collinear density-matrix propagation with a time step of 1.0 atomic unit of time and production trajectories of 40000 time steps, corresponding to approximately 1.0 ps.

Table 1: Single-defect NV⁻ supercells used in the RT-TDDFT calculations. The number of diamond sites refers to the pristine diamond supercell before creating the NV center. The actual number of atoms is one smaller after replacing one C atom by N and removing a nearest-neighbor C atom to form the vacancy.

System	Supercell	Sites	Atoms	Grid	h (a_0)	k -point sampling
NV- $x2$ (small)	$2 \times 2 \times 2$	64	63	48^3	0.281	Γ , $4 \times 4 \times 4$
NV- $x4$ (medium)	$4 \times 4 \times 4$	512	511	96^3	0.281	Γ , $2 \times 2 \times 2$
NV- $x8$ (large)	$8 \times 8 \times 8$	4096	4095	192^3	0.281	Γ

4.5.2 Polarization selection rules and spin selectivity

The polarization-resolved RT-TDDFT spectra provide a direct test of the local C_{3v} selection rules of the NV⁻ center. This analysis was performed for the medium-size single-defect NV- $x4$ supercell (511 atoms) at the Γ point. The TDDFT active space consisted of 1024 occupied and 3200 virtual orbitals (approximately 6 per each C atom), overall total 4324 orbitals. Three impulsive electric-field polarizations were considered: one parallel to the NV axis,

$$\hat{\mathbf{e}}_{\parallel} = \frac{1}{\sqrt{3}}(1, 1, 1), \quad (38)$$

and two mutually orthogonal directions perpendicular to the NV axis,

$$\hat{\mathbf{e}}_{\perp,1} = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \hat{\mathbf{e}}_{\perp,2} = \frac{1}{\sqrt{6}}(1, 1, -2). \quad (39)$$

The calculated spectra show that the defect-associated peak in the experimental zero-phonon-line energy region is absent for the polarization parallel to the $[111]$ defect axis, while it appears clearly for both transverse polarizations, $\hat{\mathbf{e}}_{\perp,1}$ and $\hat{\mathbf{e}}_{\perp,2}$. See Figure 3.

This behavior follows directly from the symmetry of the ${}^3A_2 \rightarrow {}^3E$ transition. In the local C_{3v} point group, the velocity (or dipole operator for length gauge case) component parallel to the defect axis transforms as A_1 , whereas the two transverse velocity (or dipole operator for length gauge case) components transform together as the two-dimensional E representation. The electric-dipole matrix element is symmetry allowed only when the direct product of the initial state, dipole operator, and final state contains the totally symmetric representation A_1 . For axial polarization,

$$A_2 \otimes A_1 \otimes E = E, \quad (40)$$

which does not contain A_1 , and the transition is forbidden. For transverse polarization,

$$A_2 \otimes E \otimes E = A_1 \oplus A_2 \oplus E, \quad (41)$$

which contains A_1 , and the transition is allowed. The computed polarization dependence is therefore consistent with the expected selection rule that the optical transition dipole for the $\text{NV}^- \ ^3A_2 \rightarrow \ ^3E$ transition lies perpendicular to the NV axis.

The same spectra also demonstrate spin selectivity of the defect-region optical response. In the collinear spin convention used in the present calculations, the peak in the zero-phonon-line energy region appears predominantly in the spin-down channel. Equivalently, this is the minority-spin channel of the triplet configuration shown in Fig. 2b. The dominant excitation promotes a minority-spin electron from the occupied a_1 -like defect orbital into the e manifold, producing the configuration schematically shown for the optically excited triplet state. Because the electric-dipole perturbation does not flip spin, the transition appears in the spin channel associated with this minority-spin $a_1 \rightarrow e$ excitation. This spin-resolved selection rule provides a stringent test of the spin-collinear RT-TDDFT implementation for localized defect states in a periodic host.

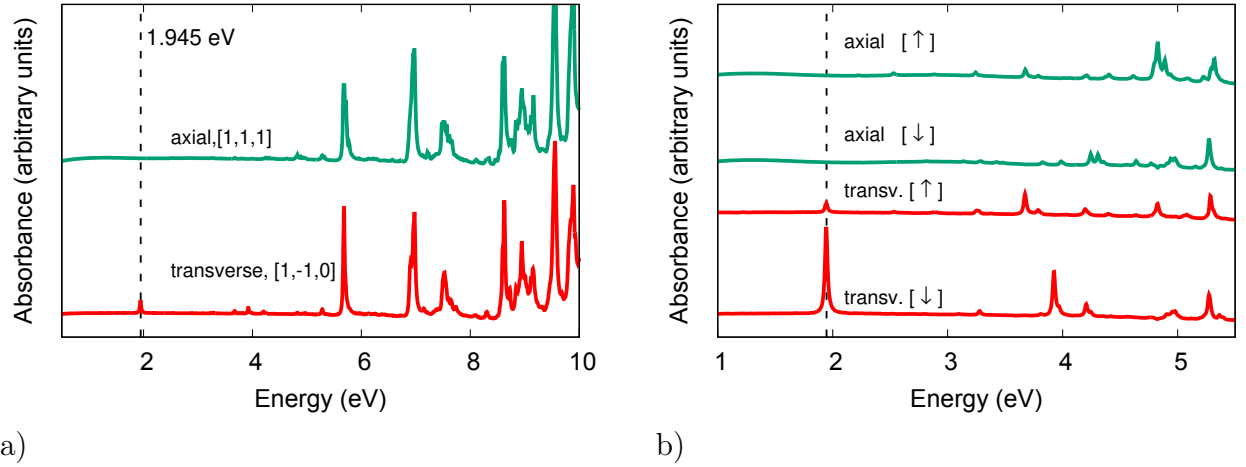


Figure 3: Polarization and spin selectivity of the NV^- defect optical response from spin-collinear RT-TDDFT for the 512-site $\text{NV-}x4$ supercell. (a) Spin-summed spectra for axial and transverse light polarizations. The axial spectrum corresponds to a kick polarized parallel to the NV axis, $\mathbf{E} \parallel [111]$, while the transverse spectrum corresponds to $\mathbf{E} \parallel [1\bar{1}0]$, perpendicular to the NV axis. The dashed vertical line marks the experimental NV^- zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. The defect-associated peak near this energy is strongly enhanced for transverse polarization and suppressed for axial polarization, consistent with the C_{3v} selection rule for the $\ ^3A_2 \rightarrow \ ^3E$ transition. (b) Expanded view of the sub-gap optical response below approximately 5.5 eV, corresponding to the transparency window of pristine diamond. In this energy range, optical features arise from the NV^- defect states rather than from intrinsic band-to-band absorption of the diamond host. The spin-resolved spectra show that the transverse response near the ZPL energy is dominated by the spin-down, minority-spin channel, consistent with the dominant $a_1 \rightarrow e$ defect excitation shown schematically in Fig. 2. The spectra are vertically shifted for clarity.

4.5.3 Active-space convergence of the defect excitation

The selection-rule analysis above used the medium-size single-defect $\text{NV-}x4$ supercell at the Γ point with the TDDFT active space including full occupied orbitals manifold and 3200 virtual orbitals. Here we use the same system and transverse polarization to examine how

the defect-associated peak depends on the active propagation manifold used in the RT-TDDFT calculation. As the system size increases, the number of occupied and unoccupied Kohn–Sham states included in the density-matrix propagation can grow rapidly, making the computational cost prohibitive. Similarly, one might expect the low-energy defect excitation to be only weakly sensitive to deep occupied states or to high-energy virtual states, since the relevant NV^- defect states are localized in the diamond band gap. We investigate the effect of the TDDFT active space construction on position of the experimental zero phonon line 1.945 eV peak to quantify the effects of this effect numerically in terms of fraction of the occupied orbitals, $f_{\text{occ}} = \frac{N_{\text{occ}}^{\text{act}}}{N_{\text{occ}}^{\text{full}}}$, and the number of virtual states per diamond site, $n_{\text{virt}} = \frac{N_{\text{virt}}}{N_{\text{site}}}$, used to construct the TDDFT active space. Here, $N_{\text{occ}}^{\text{full}}$ represents the total number of occupied orbitals, $N_{\text{occ}}^{\text{act}}$ is the number of occupied states retained in the propagation, N_{virt} is the number of unoccupied states included in the active manifold and N_{site} is the total number of diamond lattice sites before formation of the defect.

For the 512-site $NV-x4$ supercell used in this convergence test, $N_{\text{occ}}^{\text{full}} = 1024$ and $N_{\text{site}} = 512$. The occupied-space tests removed the lowest 0, 200, 400, 512, 600, and 800 occupied Kohn–Sham states from the active propagation manifold. This corresponds to retaining 1024, 824, 624, 512, 424, and 224 occupied states, or approximately 100%, 80%, 60%, 50%, 40%, and 20% of the occupied manifold, respectively. The virtual-space tests used $N_{\text{virt}} = 512$, 1024, 2048, and 3200, corresponding to $n_{\text{virt}} = 1$, 2, 4, and 6.25 virtual states per diamond site.

Figure 4 shows the effect of occupied- and virtual-space truncation on the position of the defect-associated peak near the experimental ZPL energy. Reducing f_{occ} shifts the peak to higher energy, showing that the deep occupied manifold cannot be discarded without changing the computed defect excitation. Conversely, increasing n_{virt} systematically shifts the peak to lower energy, with the peak position approaching a saturated value as the virtual space is enlarged. Physically, this behavior indicates that the localized defect excitation is dressed by the polarizable diamond environment and by the self-consistent time-dependent response of the electron density. Truncating either the occupied or unoccupied part of the propagation manifold suppresses electronic relaxation and screening around the particle-hole excitation, producing a measurable blueshift

Furthermore, the two convergence tests show different sensitivities. At fixed $n_{\text{virt}} = 6.25$, reducing the retained occupied fraction from 100% to about 50% produces only a modest blue shift of the defect-associated peak, while retaining only about 20% of the occupied space leads to a larger shift. By contrast, at fixed $f_{\text{occ}} = 100\%$, increasing the virtual active space from $n_{\text{virt}} = 1$ to 4–6.25 virtual states per site produces a substantially larger red shift and brings the peak close to the experimental ZPL energy. Thus, for this $NV-x4$ test case, the unoccupied portion of the active space is the primary convergence bottleneck for the peak position, while the occupied manifold can be moderately truncated with a smaller, but still measurable, effect.

In summary, the peak position is more sensitive to the size of the virtual active space than to moderate truncation of the occupied manifold. However, aggressive occupied-space truncation also produces a measurable blue shift, indicating that the response of deeper valence states contributes to the screening and relaxation of the localized defect excitation.

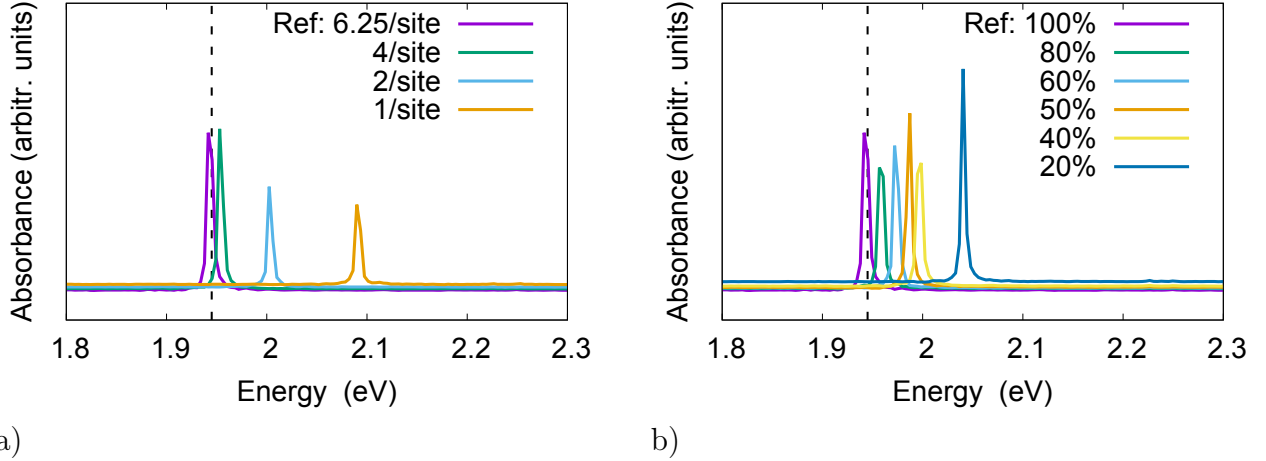


Figure 4: Active-space convergence of the defect-associated optical response for the 512-site NV- x 4 supercell at the Γ point, using transverse polarization along $[1\bar{1}0]$ direction. The dashed vertical line marks the experimental NV⁻ zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. The reference calculation, shown in both panels, uses the full occupied space, $f_{\text{occ}} = 100\%$, and $N_{\text{virt}} = 3200$, corresponding to $n_{\text{virt}} = 6.25$ virtual states per diamond site for the 512-site cell. (a) Dependence on the virtual active space, expressed as the number of virtual states per diamond site, $n_{\text{virt}} = N_{\text{virt}}/N_{\text{site}}$. The values $n_{\text{virt}} = 1, 2, 4$, and 6.25 correspond to $N_{\text{virt}} = 512, 1024, 2048$, and 3200 , respectively. Increasing n_{virt} systematically red-shifts the peak toward a saturated value. (b) Dependence on the retained occupied fraction, $f_{\text{occ}} = N_{\text{occ}}^{\text{act}}/N_{\text{occ}}^{\text{full}}$. The retained values 100%, 80%, 60%, 50%, 40%, and 20% are approximate and correspond to 1024, 800, 600, 512, 400, and 200 occupied states. Reducing f_{occ} shifts the defect-associated peak to higher energy.

4.5.4 Defect concentration and magnetic alignment in replicated NV⁻ arrays

The large NV- x 8 supercell also provides a controlled framework for examining how the optical response evolves with defect concentration. Starting from the same 4096-site diamond host, we constructed cells containing 1, 8, and 64 NV⁻ centers. These systems correspond, respectively, to the defect concentrations obtained from the NV- x 8, NV- x 4, and NV- x 2 cells: the 8-NV structure is generated by a $2 \times 2 \times 2$ replication of the NV- x 4 cell, whereas the 64-NV structure is generated by a $4 \times 4 \times 4$ replication of the NV- x 2 cell. The comparison therefore changes the number and separation of the defects while retaining the same total supercell size, real-space grid, transverse field polarization, spectral broadening, and Γ -point representation. The concentration series discussed first uses a parallel collinear alignment of the NV moments.

The RT-TDDFT calculations for the three 4096-site systems also used the same active-space density. Approximately 67% of the occupied space was retained, while the virtual space contained approximately two unoccupied states per diamond site. In the RMG state-count convention, the calculations used 16400 computed states, with 2700 deep occupied states frozen and four highest-energy states discarded, leaving 13696 states in the density-matrix propagation. Each spectrum was obtained from 42000 electronic time steps with $\Delta t = 1.0$ a.u., corresponding to a total propagation time of approximately 1.02 ps. Consequently, differences among the three RT-TDDFT spectra predominantly reflect the NV concentration rather than changes in total cell size, propagation length, or active-space density.

For comparison, we also evaluated the optical response within the independent-particle

approximation (IPA) using the LOPTICS implementation in RMG. In atomic units, the spin-resolved imaginary part of the IPA dielectric tensor can be written as

$$\epsilon_{\alpha\beta}^{(2),\sigma}(\omega) = \frac{4\pi^2}{\Omega\omega^2} \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{v,c} (f_{v\mathbf{k}\sigma} - f_{c\mathbf{k}\sigma}) \text{Re} \left[P_{v\mathbf{k}\sigma}^{\alpha*} P_{c\mathbf{k}\sigma}^{\beta} \right] \delta_{\eta} (\epsilon_{c\mathbf{k}\sigma} - \epsilon_{v\mathbf{k}\sigma} - \omega), \quad (42)$$

where Ω is the supercell volume, $w_{\mathbf{k}}$ is the $\mathbf{k}$ -point weight, and $P_{v\mathbf{k}\sigma}^{\alpha}$ is the velocity-gauge transition matrix element, including the nonlocal-pseudopotential contribution. The indices v and c denote occupied and unoccupied Kohn–Sham states of the same global spin σ , and δ_{η} represents the broadened energy-conservation condition. The IPA spectra therefore expose the underlying vertical Kohn–Sham transition structure, whereas the RT-TDDFT spectra additionally contain the self-consistent induced Hartree and adiabatic exchange–correlation response. The IPA curves in Fig. 5 were independently scaled for clarity, so the comparison concerns peak positions and spectral structure rather than absolute intensities.

Figure 5 shows a pronounced evolution of the optical response with decreasing NV concentration. For the 64-NV array, corresponding to the NV- $x2$ concentration, both IPA and RT-TDDFT produce strongly fragmented spectra over the sub-gap energy range. The large number of optically active features is consistent with substantial coupling among the closely spaced defects, which hybridizes the defect-derived states and produces dispersive occupied and unoccupied defect bands. The NV- $x2$ concentration is therefore clearly outside the isolated-defect regime.

At the intermediate NV- $x4$ concentration, represented by eight defects in the same 4096-site host, the spectral response becomes much simpler. The IPA spectrum retains several closely spaced transitions in the vicinity of the experimental ZPL energy, indicating that some fine structure remains in the underlying Kohn–Sham transition spectrum. By contrast, the self-consistent RT-TDDFT response is dominated by a single resonance near 2.01 eV. This simplification is consistent with coupling and redistribution of oscillator strength among nearby independent-particle transitions in the time-dependent response. Finite spectral resolution and the truncated RT-TDDFT active space may also suppress weaker components, so the single RT-TDDFT peak should not be interpreted as evidence that every periodic-image effect has disappeared. Nevertheless, it shows that the dominant optical observable is already substantially single-resonance-like at the NV- $x4$ defect separation.

Further dilution to a single NV[−] center in the 4096-site NV- $x8$ cell produces a similarly simple low-energy response. The qualitative similarity of the dominant RT-TDDFT peaks for the 8-NV and 1-NV systems supports the conclusion that a separation corresponding to the 512-site NV- $x4$ model is sufficient to recover the principal defect-associated optical feature within the present spectral resolution and active-space approximation. The calculated peak near 2.01 eV remains approximately 0.06 eV above the experimental 1.945-eV ZPL. As shown by the active-space analysis in Sec. ??, part of this residual blueshift is consistent with the restricted occupied and virtual propagation spaces used for the 4096-site calculations. The experimental ZPL is nevertheless used only as a reference because the present frozen-ion adiabatic-PBE response also omits excited-state structural relaxation and vibronic effects.

The fixed-volume replicated cells also permit examination of how the magnetic arrangement of the NV moments affects the spin-resolved optical response. We considered both a parallel collinear configuration, denoted FM below, and a G-type antiparallel broken-

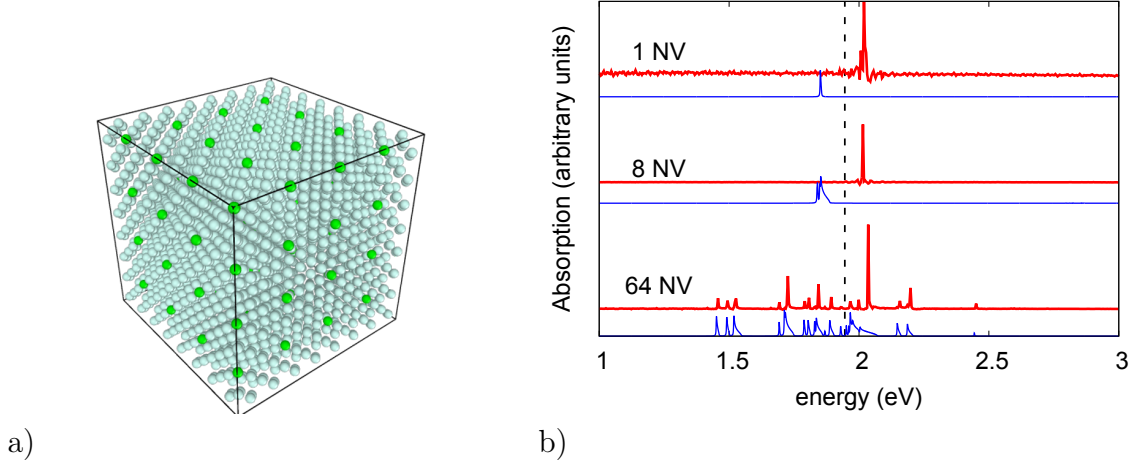


Figure 5: Concentration dependence of the optical response in parallel-aligned NV^- arrays. Spin-collinear RT-TDDFT spectra are shown in red and independent-particle approximation (IPA) spectra, evaluated with the `LOPTICS` implementation in `RMG`, are shown in blue. From top to bottom, the systems contain 1, 8, and 64 NV^- centers in the same 4096-site diamond host, corresponding to the $\text{NV-}x8$, $\text{NV-}x4$, and $\text{NV-}x2$ defect concentrations, respectively. All calculations use the same atomic structures, real-space grid, Γ -point representation, transverse polarization, and spectral broadening. The RT-TDDFT calculations additionally use the same retained occupied fraction and number of virtual states per diamond site and were propagated for 42000 steps with $\Delta t = 1.0$ a.u., corresponding to approximately 1.02 ps. The dashed vertical line marks the experimental zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. The strongly fragmented spectrum of the 64-NV array reflects substantial coupling among closely spaced defects. At the 8-NV concentration, the IPA retains nearby independent-particle transition structure, whereas the RT-TDDFT response is dominated by a single resonance resembling that of the dilute single-NV system. The spectra are vertically offset, and the IPA curves are independently scaled, for clarity.

symmetry configuration, denoted G-AFM, for the 64-NV and 8-NV arrays. In the FM configuration, all local defect moments are aligned with the same global spin axis. In the G-AFM configuration, the local moment orientation alternates between neighboring defect sites.

The persistence of the local moments in these configurations can be characterized using the total and absolute magnetizations,

$$M_{\text{tot}} = \mu_B \int [\rho_{\uparrow}(\mathbf{r}) - \rho_{\downarrow}(\mathbf{r})] d\mathbf{r}, \quad M_{\text{abs}} = \mu_B \int |\rho_{\uparrow}(\mathbf{r}) - \rho_{\downarrow}(\mathbf{r})| d\mathbf{r}. \quad (43)$$

For the 8-NV array, the FM solution gives $M_{\text{tot}} = 20.73 \mu_B$ and $M_{\text{abs}} = 22.36 \mu_B$, whereas the G-AFM solution gives $M_{\text{tot}} = 0$ and $M_{\text{abs}} = 18.66 \mu_B$. The corresponding values for the 64-NV array are $M_{\text{tot}} = 128.42 \mu_B$ and $M_{\text{abs}} = 143.03 \mu_B$ for the FM arrangement, and $M_{\text{tot}} = 0$ and $M_{\text{abs}} = 138.57 \mu_B$ for the G-AFM arrangement. The vanishing total magnetization but substantial absolute magnetization of the G-AFM solutions confirms that the local defect moments remain present and cancel globally rather than disappearing.

The spin-resolved spectra in Fig. 6 show how the local minority-spin selection rule established for the single NV center maps onto the globally defined spin channels of a multi-defect array. In the FM configuration, all local moments have the same orientation. The local minority-spin $a_1 \rightarrow e$ excitation therefore corresponds to the same global spin channel at

every defect, and the optical response is strongly dominated by one spin component. In the G-AFM arrangement, the local moment is reversed on the second magnetic sublattice. The local minority-spin excitation consequently maps onto one global spin channel on one sublattice and the opposite global spin channel on the other. The corresponding defect-associated features therefore appear in both global spin-resolved spectra.

This redistribution between global spin channels does not represent a spin-flip optical transition. The optical perturbation remains spin independent and the transitions conserve the global collinear spin label. Instead, the result reflects the reversal of the local moment relative to the fixed global quantization axis. Local minority spin is represented by opposite global spin labels on the two G-AFM sublattices.

The concentration dependence found above is also evident in the magnetic comparison. At the NV- $x2$ concentration, both FM and G-AFM spectra contain many defect-associated peaks, and the detailed distribution of these features depends on the magnetic arrangement. At the more dilute NV- $x4$ concentration, the response is concentrated primarily in a single resonance near 2.0 eV. Changing from FM to G-AFM then affects mainly the distribution of this resonance between the two global spin channels rather than producing a comparably strong fragmentation of the spectrum. This behavior is consistent with a substantial reduction of electronic coupling among the NV centers as their separation increases.

Together, these calculations show that increasing the NV–NV separation transforms the optical response from the fragmented spectrum of an interacting defect array to a dominant isolated-defect-like resonance. At the same time, the magnetic alignment determines how the local minority-spin excitation is represented in the globally defined spin channels. The 4096-site calculations therefore demonstrate that the periodic spin-collinear RT-TDDFT implementation can treat both dilute defects and extended arrays containing as many as 64 optically active defect centers within the same real-space simulation framework.

References

- (1) Marques, M. A. L., Maitra, N. T., Nogueira, F. M. S., Gross, E. K. U., Rubio, A., Eds. *Fundamentals of Time-Dependent Density Functional Theory*; Lecture Notes in Physics; Springer: Heidelberg, Germany, 2012; Vol. 837.
- (2) Li, X.; Govind, N.; Isborn, C.; A. Eugene DePrince, I.; ; Lopata, K. Real-Time Time-Dependent Electronic Structure Theory. *Chem. Rev.* **2020**, *120*, 9951 – 9993.
- (3) Lopata, K.; Govind, N. Modeling Fast Electron Dynamics with Real-time Time-dependent Density Functional Theory: Application to Small Molecules and Chromophores. *J. Chem. Theory Comput.* **2011**, *7*, 1344 – 1355.
- (4) Silverstein, D. W.; Govind, N.; Van Dam, H. J.; Jensen, L. Simulating One-Photon Absorption and Resonance Raman Scattering Spectra using Analytical Excited State Energy Gradients within Time- Dependent Density Functional Theory. *J. Chem. Theory Comput.* **2013**, *9*, 5490–5503.
- (5) Lee, K.-M.; Yabana, K.; Bertsch, G. Magnetic Circular Dichroism in Real-time Time-dependent Density Functional Theory. *J. Chem. Phys.* **2011**, *134*, 144106.

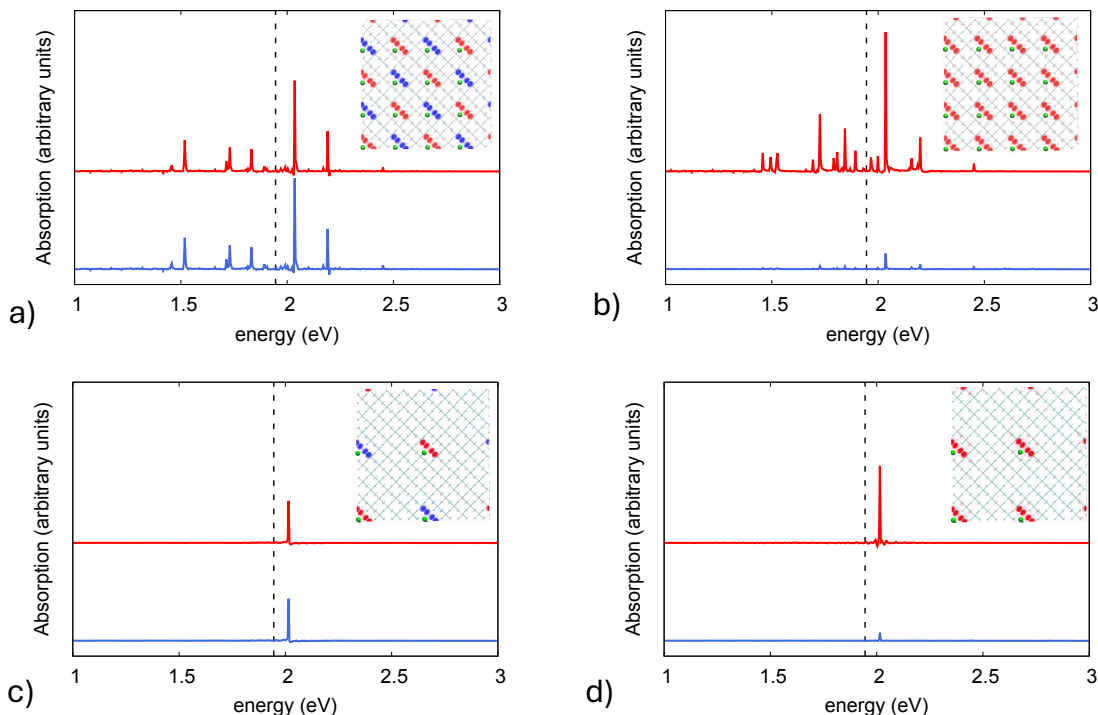


Figure 6: Spin-resolved RT-TDDFT optical response of replicated NV^- arrays in the 4096-site diamond host. The upper row shows the 64-NV system at the $\text{NV}-x2$ concentration, and the lower row shows the 8-NV system at the $\text{NV}-x4$ concentration. Panels (a) and (c) correspond to G-type antiparallel broken-symmetry arrangements, whereas panels (b) and (d) correspond to parallel, FM arrangements. The red and blue curves show the two globally defined collinear spin channels and are vertically offset for clarity. The insets illustrate the corresponding local defect-spin arrangements; red and blue markers denote opposite local moment orientations. The dashed vertical lines mark the experimental NV^- zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. In the FM configurations, the local minority-spin defect excitation maps predominantly onto one global spin channel. In the G-AFM configurations, reversal of the local moments between the two magnetic sublattices causes the same local minority-spin excitation to appear in both global spin channels. The response remains strongly fragmented at the $\text{NV}-x2$ concentration but becomes predominantly single-resonance-like at the $\text{NV}-x4$ concentration.

- (6) De Giovannini, U.; Hübener, H.; Rubio, A. Time-dependent Density Functional Theory Framework for Spin and Time-resolved Angular-resolved Photoelectron Spectroscopy in Periodic Systems. *J. Chem. Theory Comput.* **2017**, *13*, 265–273.
- (7) Herbert, J. M.; Zhu, Y.; Alam, B.; Ojha, A. K. Time-Dependent Density Functional Theory for X-ray Absorption Spectra: Comparing the Real-Time Approach to Linear Response. *J. Chem. Theory Comput.* **2023**, *19*, 6745 – 6760.
- (8) Bhat, V.; Callaway, C. P.; Risko, C. Computational Approaches for Organic Semiconductors: From Chemical and Physical Understanding to Predicting New Materials. *Chem. Rev.* **2023**, *123*, 7498–7547.
- (9) Jakowski, J.; Irle, S.; Sumpter, B.; Morokuma, K. Modeling charge transfer in fullerene collisions via real-time electron dynamics. *J. Phys. Chem. Lett* **2012**, *3*, 1536–1542.

- (10) Zeng, S.; Baillargeat, D.; Ho, H.-P.; Yong, K.-T. Nanomaterials enhanced surface plasmon resonance for biological and chemical sensing applications. *Chem. Soc. Rev.* **2014**, *43*, 3426–3452.
- (11) Ilawe, N. V.; Oviedo, M. B.; Wong, B. M. Dynamics of Long-Range Electronic Excitation Transfer in Plasmonic Nanoantennas. *J. Chem. Theory Comput.* **2017**, *13*, 3442–3454.
- (12) Yu, T.; Lingerfelt, D.; Jakowski, J.; Ganesh, P.; Sumpter, B. G. Electron-Beam Induced Molecular Plasmon Excitations and Energy Transfer in Silver Molecular Nanowires. *J. Phys. Chem. A* **2021**, *125*, 74–87.
- (13) Ding, F.; Goings, J. J.; Frisch, M. J.; Li, X. Ab Initio Non-Relativistic Spin Dynamics. *J. Chem. Phys.* **2014**, *141*, 214111.
- (14) Goings, J. J.; Kasper, J. M.; Egidi, F.; Sun, S.; Li, X. Real Time Propagation of the Exact Two Component Time-Dependent Density Functional Theory. *J. Chem. Phys.* **2016**, *145*, 104107.
- (15) Peralta, J. E.; Hod, O.; Scuseria, G. E. Magnetization Dynamics from Time-Dependent Noncollinear Spin Density Functional Theory Calculations. *J. Chem. Theory Comput.* **2015**, *11*, 3661 – 3668.
- (16) Jakowski, J.; Lu, W.; Briggs, E.; Lingerfelt, D.; Sumpter, B. G.; Ganesh, P.; Bernholc, J. Simulation of 24,000 Electron Dynamics: Real-Time Time-Dependent Density Functional Theory (TDDFT) with the Real-Space Multigrids (RMG). *Journal of Chemical Theory and Computation* **2025**, *21*, 1322–1339.
- (17) Castro, A.; Appel, H.; Oliveira, M.; Rozzi, C. A.; Andrade, X.; Lorenzen, F.; Marques, M. A. L.; Gross, E. K. U.; Rubio, A. Octopus: a Tool for the Application of Time-dependent Density Functional. *Theory. Phys. Status Solidi B* **2006**, *243*, 2465–2488.
- (18) W.Draeger, E.; Andrade, X.; Gunnels, J. A.; Bhatele, A.; Schleife, A.; A.Correa, A. Massively parallel first-principles simulation of electron dynamics in materials. *Journal of Parallel and Distributed Computing* **2017**, *106*, 205–214.
- (19) Shepard, C.; Ruiyi Zhou, D. C. Y.; Yao, Y.; Kanai, Y. Simulating electronic excitation and dynamics with real-time propagation approach to TDDFT within plane-wave pseudopotential formulation. *J. Chem. Phys.* **2021**, *155*, 100901.
- (20) Kühne, T.; Iannuzzi, M.; Ben, M. D.; Rybkin, V. V.; Seewald, P.; Stein, F.; Laino, T.; Khaliullin, R. Z.; Schütt, O.; Schiffmann, F. et al. CP2K: An electronic structure and molecular dynamics software package - Quickstep: Efficient and accurate electronic structure calculations. *J. Chem. Phys.* **2020**, *150*, 194103.
- (21) Takimoto, Y.; Vila, F. D.; Rehr, J. J. Real-time time-dependent density functional theory approach for frequency-dependent nonlinear optical response in photonic molecules. *J. Chem. Phys.* **2007**, *127*, 154114.

- (22) Hourahine, B.; Aradi, B.; V. Blum, e. a. DFTB+, a software package for efficient approximate density functional theory based atomistic simulations. *J. Chem. Phys.* **2020**, *152*, 124101.
- (23) Rufus, N. D.; Kanungo, B.; Gavini, V. Fast and robust all-electron density functional theory calculations in solids using orthogonalized enriched finite elements. *Phys. Rev. B* **2021**, *104*, 085112.
- (24) Briggs, E. L.; Lu, W.; Bernholc, J. Adaptive finite differencing in high accuracy electronic structure calculations. *npj Comput. Mater.* **2024**, *10*, 17.
- (25) Mattiat, J.; Luber, S. Comparison of Length, Velocity, and Symmetric Gauges for the Calculation of Absorption and Electric Circular Dichroism Spectra with Real-Time Time-Dependent Density Functional Theory. *J. Chem. Theory Comput.* **2022**, *18*, 5513 – 5526.
- (26) Kleinman, L.; Bylander, D. M. Efficacious Form for Model Pseudopotentials. *Phys. Rev. Lett.* **1982**, *48*, 1425.
- (27) Vanderbilt, D. Soft self-consistent pseudopotentials in a generalized eigenvalue formalism. *Phys. Rev. B* **1990**, *41*, 7892.
- (28) Doherty, M. W.; Manson, N. B.; Delaney, P.; Jelezko, F.; Wrachtrup, J.; Hollenberg, L. C. L. The nitrogen-vacancy colour centre in diamond. *Phys. Rep.* **2013**, *528*, 1–45.
- (29) Gali, Á.; Fyta, M.; Kaxiras, E. Ab initio supercell calculations on nitrogen-vacancy center in diamond: Electronic structure and hyperfine tensors. *Phys. Rev. B* **2008**, *77*, 155206.
- (30) Gali, Á. Ab initio theory of the nitrogen-vacancy center in diamond. *Nanophotonics* **2019**, *8*, 1907–1943.
- (31) Choi, S.; Jain, M.; Louie, S. G. Mechanism for optical initialization of spin in NV[−] center in diamond. *Phys. Rev. B* **2012**, *86*, 041202(R).
- (32) Balasubramanian, G.; Neumann, P.; Twitchen, D.; Markham, M.; Kolesov, R.; Mizuochi, N.; Isoya, J.; Achard, J.; Beck, J.; Tissler, J. et al. Ultralong spin coherence time in isotopically engineered diamond. *Nat. Mater.* **2009**, *8*, 383–387.
- (33) Degen, C. L.; Reinhard, F.; Cappellaro, P. Quantum sensing. *Rev. Mod. Phys.* **2017**, *89*, 035002.
- (34) Maze, J. R.; Stanwix, P. L.; Hodges, J. S.; Hong, S.; Taylor, J. M.; Cappellaro, P.; Jiang, L.; Gurudev Dutt, M. V.; Togan, E.; Zibrov, A. S. et al. Nanoscale magnetic sensing with an individual electronic spin in diamond. *Nature* **2008**, *455*, 644–647.
- (35) Schirhagl, R.; Chang, K.; Loretz, M.; Degen, C. L. Nitrogen-vacancy centers in diamond: nanoscale sensors for physics and biology. *Annu. Rev. Phys. Chem.* **2014**, *65*, 83–105.

- (36) Aharonovich, I.; Englund, D.; Toth, M. Solid-state single-photon emitters. *Nat. Photonics* **2016**, *10*, 631–641.
- (37) Bradley, C. E.; Randall, J.; Abobeih, M. H.; Berrevoets, R. C.; Degen, M. J.; Bakker, M. A.; Markham, M.; Twitchen, D. J.; Taminiau, T. H. A ten-qubit solid-state spin register with quantum memory up to one minute. *Phys. Rev. X* **2019**, *9*, 031045.